

4-1 Quadratic Functions and Transformations

Algebra 2 · Chapter 4: Quadratic Functions and Equations

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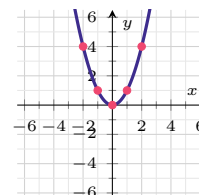
I can: recognize the parent function $y = x^2$ · describe how a , h , k transform the graph · find the vertex, axis, max/min, domain and range from vertex form · write a quadratic function from a graph or a description.

Vocabulary: parabola, vertex, axis of symmetry, vertex form, minimum/maximum value, reflection, translation, vertical stretch, vertical compression

1. Warm-Up: The Parent Function $f(x) = x^2$

Complete the table, then plot the points.

x	-3	-2	-1	0	1	2	3
$f(x)$	<u>9</u>	<u>4</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>4</u>	<u>9</u>



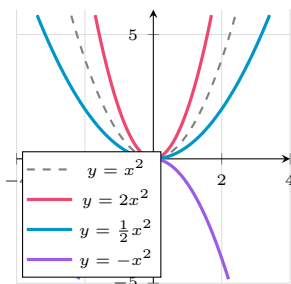
Notice

The outputs for x and $-x$ are the same, so the graph is symmetric.

Definition

A **quadratic function** can be written as $f(x) = ax^2 + bx + c$, where $a \neq 0$. Its graph is a U-shaped curve called a **parabola**. The **vertex** is the **turning point**; the **axis of symmetry** is the **vertical line** through the vertex. For $f(x) = x^2$: vertex (0, 0), axis $x = 0$.

2. What Does a Do? $f(x) = ax^2$



Notice

- $y = 2x^2$ is narrower than $y = x^2$ (each y is doubled).
- $y = \frac{1}{2}x^2$ is wider than $y = x^2$.
- $y = -x^2$ opens down: a reflection in the x -axis.
- The vertex stays at (0, 0).

Value of a	Effect on $y = x^2$
$ a > 1$	vertical <u>stretch</u> (narrower)
$0 < a < 1$	vertical <u>compression</u> (wider)
$a > 0$	opens <u>up</u> ; vertex is a <u>minimum</u>
$a < 0$	opens <u>down</u> ; vertex is a <u>maximum</u>

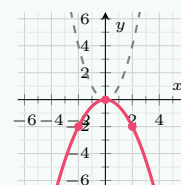
Common Mistake

Stretch/compression depends on $|a|$, not the sign: $y = -3x^2$ is a reflection *and* a vertical stretch.

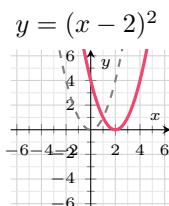
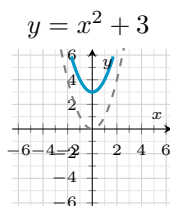
Watch Me 1 — Graph $f(x) = -\frac{1}{2}x^2$

x	-4	-2	0	2	4
$f(x)$	<u>-8</u>	<u>-2</u>	<u>0</u>	<u>-2</u>	<u>-8</u>

Compared with $y = x^2$: reflection in the x -axis; vertical compression by a factor of $\frac{1}{2}$; vertex (0, 0) is a maximum.



3. What Do h and k Do?



Notice

$y = x^2 + 3$ moves the graph **up** 3.

$y = (x - 2)^2$ moves the graph **right** 2.

$x - 2$ *inside* the square moves the graph the **opposite** way of the sign.

Function	Translation of $y = x^2$	Vertex
$y = x^2 + k$	up if $k > 0$, down if $k < 0$	<u>$(0, k)$</u>
$y = (x - h)^2$	right if $h > 0$, left if $h < 0$	<u>$(h, 0)$</u>
$y = (x + 3)^2$	left <u>3</u> (here $h = -3$)	<u>$(-3, 0)$</u>

Common Mistake

$(x + 3)^2 = (x - (-3))^2$, so $h = -3$: the graph moves **left**, not right.

4. Vertex Form

Vertex Form $f(x) = a(x - h)^2 + k$

- vertex: (h, k) axis: $x = h$
- $a > 0$: opens up, min value k
- $a < 0$: opens down, max value k
- domain: **all real numbers**
- range: $y \geq k$ if $a > 0$; $y \leq k$ if $a < 0$
- $|a|$ controls **stretch / compression**

Order to graph: **reflect** /stretch with $a \rightarrow$ shift **horizontally** by $h \rightarrow$ shift **vertically** by k

Watch Me 2 — $f(x) = (x + 3)^2 - 4$

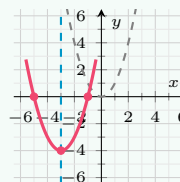
Rewrite: $f(x) = 1(x - (-3))^2 + (-4)$

$a =$ 1, $h =$ -3, $k =$ -4

Transformations: **left** 3, **down** 4

Vertex $(-3, -4)$ Axis $x = -3$ **Minimum** value -4

Domain: **all real numbers** Range: $y \geq -4$

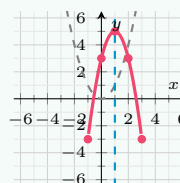


Watch Me 3 — $f(x) = -2(x - 1)^2 + 5$

$a =$ -2, $h =$ 1, $k =$ 5

Transformations (in order): **reflect in x-axis**, vertical **stretch** by 2, **right** 1, **up** 5

Vertex $(1, 5)$ Axis $x = 1$ **Maximum** value 5 Range: $y \leq 5$

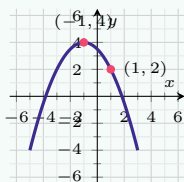


Your Turn — Read the Vertex Form

Function	Vertex	Axis	Max or Min?	Value
$f(x) = (x - 5)^2 + 2$	<u>$(5, 2)$</u>	<u>$x = 5$</u>	<u>min</u>	<u>2</u>
$g(x) = -(x + 1)^2 - 3$	<u>$(-1, -3)$</u>	<u>$x = -1$</u>	<u>max</u>	<u>-3</u>
$p(x) = \frac{1}{2}x^2 - 6$	<u>$(0, -6)$</u>	<u>$x = 0$</u>	<u>min</u>	<u>-6</u>

5. Writing Equations

Watch Me 4 — From a Graph



Step 1. Vertex $(h, k) = (-1, 4)$, so $f(x) = a(x + 1)^2 + 4$

Step 2. Use the point $(1, 2)$: $2 = a(1 + 1)^2 + 4 \Rightarrow 2 = 4a + 4 \Rightarrow a = -\frac{1}{2}$

Step 3. $f(x) = -\frac{1}{2}(x + 1)^2 + 4$

Check: it opens down, so a should be negative ✓

Watch Me 5 — From a Description

Start with $y = x^2$. Reflect it in the x -axis, stretch it vertically by 3, then translate it **4 units left** and **1 unit down**.

reflect + stretch by 3 $\Rightarrow a = -3$ 4 left $\Rightarrow h = -4$ 1 down $\Rightarrow k = -1$ $g(x) = -3(x + 4)^2 - 1$

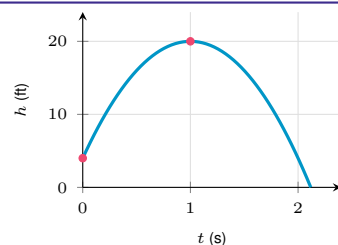
What If...?

- a changes from 2 to -2 . Does the vertex move? No. Only the direction changes.
- Swap h and k : $(x - 2)^2 + 5$ vs. $(x - 5)^2 + 2$? Different vertices: $(2, 5)$ vs. $(5, 2)$.
- $a = 0$: is $y = 0(x - 1)^2 + 3$ a parabola? No — it is the line $y = 3$.

6. Connection: A Water Fountain

The height (feet) of a water stream t seconds after it leaves the fountain is $h(t) = -16(t - 1)^2 + 20$.

- Maximum height: 20 ft It reaches the top after 1 second.
- Starting height $h(0) =$ 4 ft
- Why is the vertex a maximum? $a = -16 < 0$, so it opens down.



7. Practice

Practice

- Describe the transformations of $y = x^2$: $g(x) = 3(x - 2)^2 + 1$
vertical stretch by 3, right 2, up 1
- Describe the transformations of $y = x^2$: $g(x) = -\frac{1}{4}(x + 5)^2$
reflect in x -axis, vertical compression by $\frac{1}{4}$, left 5
- Write the function: $y = x^2$ reflected in the x -axis, then 6 right and 2 up. $g(x) = -(x - 6)^2 + 2$
- Vertex $(3, 1)$; the graph passes through $(5, 9)$. $f(x) = 2(x - 3)^2 + 1$
 $9 = a(5 - 3)^2 + 1 \Rightarrow 4a = 8 \Rightarrow a = 2$
- Challenge.** $f(x) = a(x - h)^2 + k$ passes through $(0, 5)$ and $(4, 5)$, and its minimum value is -3 . Find $f(x)$.
Same height at $x = 0$ and $x = 4 \Rightarrow$ axis $x = 2$; vertex $(2, -3)$; $5 = a(0 - 2)^2 - 3 \Rightarrow a = 2$.
 $f(x) = 2(x - 2)^2 - 3$

8. Summary & Exit Ticket

Summary

Parent function	$y = x^2$, vertex <u>$(0, 0)$</u> , axis <u>$x = 0$</u>
Vertex form	$f(x) = a(x - h)^2 + k$, vertex <u>(h, k)</u>
a	sign: opens up/down; $ a $: stretch or compression
h	horizontal shift — the opposite of the sign in $(x - h)$
k	vertical shift; also the max or min value

Exit Ticket

For $f(x) = -3(x + 2)^2 + 7$: vertex $(-2, 7)$, axis $x = -2$, opens **down**, **maximum** value **7**, range $y \leq 7$.

Transformations of $y = x^2$: **reflect in x -axis, vertical stretch by 3, left 2, up 7**

More practice: fangmath.com → Algebra 2 → Chapter 4 → Interactive practice (Quadratic Transformations).